

NHV (Noise, Harshness, Vibration) Power BI Dashboard

- Sai Ganesh Abburi

Approach Overview:

Sourcing Data: (Extract)

For this task, I spent a fair amount of time collecting data, organizing my data files and just trying to figure out what I am missing and what else do I need. The most important thing that I have learned from my academic and professional work experience is that having quality data makes the life that much easier. I publicly sourced datasets for noise, cars and harshness customer feedback reviews (sources cited in references). I could not find Vibration data publicly available so I synthesized my own Vibration data which closely follows ISO 2631-1, RMS acceleration is calibrated from IRI (road roughness) and speed. A component table has been generated as well, just 5 rows of different components to provide a richer feature distribution. 38% of noise recordings were synthetically filled as well based on the pattern from the existing data rows to get a decent coverage of noise for all Models of the Cars.

Cleaning and Preparation: (Transform)

Data cleaning was done in Power Query after importing all the tables. All the cleaning steps will be listed in the Power Query for every table. Important cleaning steps done: dealt with missing data by deleting rows with missing data which didn't matter into the overall distribution, to where rows were important: the missing data was imputed with the appropriate value. (Ex: null in review rating – to 0, null in review – to “ No review”) Performed Joins across tables, Removed duplicates, Unpivoted raw noise columns, Fixed formats across data types, fixed trailing white spaces in text, Created date table with correct format and marked as Date. After the transformation was completed, it was time for data modeling and creating the actual Measures and Columns that will help build our dashboard visualizations.

Building the model and dashboard: (After Load)

All the relationships were checked, verified relationship directionality and cardinality. Three dimension tables (Vehicle, Component, Date) joined to three fact tables (Noise, Vibration, Harshness) across eight relationships, with about thirty DAX measures. All Measures and Columns along with proper Hierarchies were created. A separate Measures table was also kept storing all Measures in their respective folders for readability and ease of access.

The bigger decision now was what to actually do with all that I have ready at my fingertips. After analyzing the data a bit more, what I found: every measured NVH metric improved between 2008 and 2018, noise down 3.3%, vibration down 22.5%, yet comfort ratings fell anyway and critical reviews more than doubled. That became the backbone of the whole dashboard story I wanted to focus on.

Page one states this paradox outright. Page two goes after the strongest root in the vibration data, the road surface,. Page three does the same for noise and customer sentiment and ends in

a table that drills straight through into individual customer reviews. The final short closing page distills all of the insights into something readable in under a minute.

Key Insights and Recommendations:

Noise and vibration both improved substantially, comfort ratings fell anyway, and critical reviews more than doubled. The problem was never really in the engineering. Road surface turns out to explain most of the vibration complaints. Suspension tuning for vibration should be benchmarked against matched road conditions since surface type alone explains most of the vibration variance.

Loudness and satisfaction don't track together either. Sports cars are the loudest segment in the fleet and the best-rated. SUVs are the quietest and the most criticized. Perhaps this could be a mismatch in expectations from the client, nevertheless vehicle manufactures should strive towards tuning sound to what each segment expects rather than just turning it down.

Further cuts to raw noise or vibration numbers probably won't move satisfaction much further. The bigger opportunity is closing the gap between what gets measured and what customers actually report feeling.

Assumptions and Limitations

About 38% of the noise readings are synthetic rather than directly measured, so read trends here as directional, not exact. Hard limitation I ran into was the Vibration data was not publicly available for access, thereby requiring a synthetic version of it to fill in the gap.

References:

1. Cars dataset: <https://www.kaggle.com/datasets/ljanjughazyran/cars1/data>
2. Noise dataset: <https://www.kaggle.com/datasets/murtio/car-noise-specification>,
<https://www.auto-decibel-db.com> (scraped additional rows)
3. Vibration dataset: Generated Synthetically.
4. Component dataset: Generate Synthetically.
5. Harshness Feedback dataset: github.com/ankkur13/Edmunds-Car-Consumer-Ratings-and-Reviews (filtered using python to match reviews to NHV-relevant-keywords)